University of Dhaka
Department of Criminology
Professional Masters in Criminology and Criminal Justice

Course: MCCJ 525 – Cybercrimes and Security
Assignment: Assessing Security Features of Online Platforms

Locked or Leaked?

Assessing the Security Features of Facebook, Instagram, WhatsApp, Gmail and Telegram

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Locked or Leaked? Assessing the Security Features of Five Online Platforms

1. Introduction

Social media usage and hacking rates have both increased many times over in the past decade. Every day we spend a massive amount of time scrolling and visiting social media accounts, which in turn increases our exposure and vulnerability to attackers. Losing a social media account to a hacker can cause more harm than losing a wallet. Almost 64 million Bangladeshis use Facebook (Statista, 2025). A 2024 study by the Cyber Crime Awareness Foundation (CCAF) found that account hacking was the most common cybercrime, affecting 21.65% of victims. About 59% of the victims were women, 78.78% were aged between 18 and 30, and only 12% went to the police (Dhaka Tribune, 2024).

For this assignment, I examined five of the most commonly used online platforms: Facebook, Instagram, WhatsApp, Gmail and Telegram. These were selected because of how frequently ordinary users use them every day. I checked each app on my own phone and cross-checked the features against their official help pages and news reports. An evaluation system was used to score the security features from 1 (weak) to 5 (strong) on five benchmarks: login protection, encryption by default, threat alerts, account recovery safety and privacy controls.

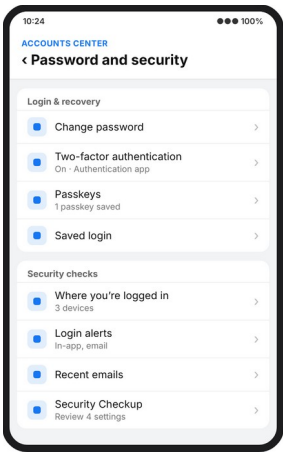
2. Existing Security Features

2.1 Facebook

Facebook's protection features sit in Settings → Accounts Center → Password and security. For login protection, Facebook has an array of options. It includes two-factor authentication (2FA), which adds a second check after the password through an authenticator app, an SMS code or a physical security key. It also sends a login alert as a warning if someone logs in from an unfamiliar device. The "Where you're logged in" feature shows every device and lets the user log out remotely. A Security Checkup feature gives a step-by-step guide through these settings. The newest feature of Facebook is passkeys, which let the user log in with a fingerprint or face scan instead of a password. Passkeys arrived in 2025 and were extended to Messenger in September 2025 (Meta, 2025).

Figure 1

Facebook → Accounts Center → Password and security



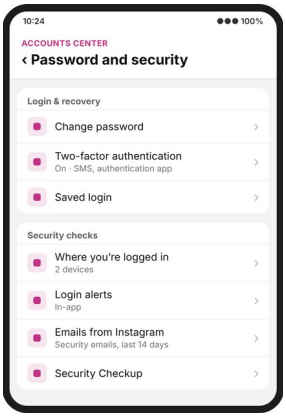
Note. Illustrative recreation of the settings screen, based on the app as of October 2026.

2.2 Instagram

Instagram shares the same Meta Accounts Center as Facebook, so its login protection features are the same. Instagram has a unique “Emails from Instagram” feature, which lists the real security emails Instagram has sent in the last 14 days. An email claiming to be from Instagram that is not on this list is likely a phishing email. It protects users from harassment through Restrict, Hidden Words and limits on message requests, which block abusive messages. Teen Accounts are private by default for users under 18. However, Meta removed its optional encrypted chats on 8 May 2026 (Euronews, 2026), which means Meta can access the content of private messages. Instagram is therefore vulnerable to sextortion and image-based blackmail, which often start in private messages.

Figure 2

Instagram → Settings → Accounts Center → Password and security



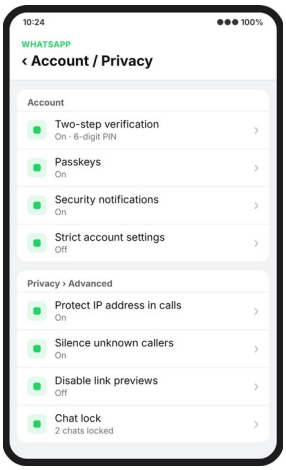
Note. Illustrative recreation of the settings screen, based on the app as of October 2026.

2.3 WhatsApp

WhatsApp encrypts every chat, call and status update end to end by default. It has extra login protection such as two-step verification, passkey login, chat lock, “Silence unknown callers” and IP protection in calls. Since October 2025, chat backups can be protected with a passkey instead of a 64-digit key (Help Net Security, 2025). In January 2026, it added Strict Account Settings, which locks everything to the safest options with one tap, blocking media from unknown senders and turning off link previews (WhatsApp, 2026). However, WhatsApp verifies a login by sending a code to the user’s phone number, and scammers exploit this. They request a code for the user’s number and then ask the user to send it back, claiming it was sent by mistake. If the user shares the code, the scammer can take over the account. A two-step PIN prevents this, but many users do not set one up.

Figure 3

WhatsApp → Settings → Account → Two-step verification / Privacy → Advanced



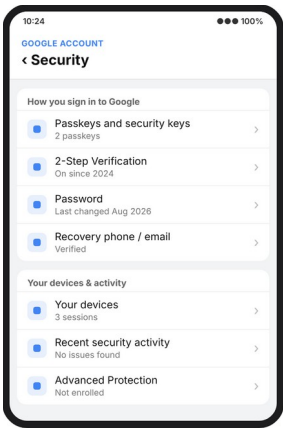
Note. Illustrative recreation of the settings screen, based on the app as of October 2026.

2.4 Gmail

Gmail’s protection is available in the Google Account (Settings → Security). Passkeys have been the default sign-in option since October 2023 (Google, 2023). Login protection also includes 2-Step Verification. Google also has an Advanced Protection Program for journalists and activists, which requires a passkey or a physical security key. According to Google, Gmail blocks more than 99.9% of phishing and malware attempts (Engadget, 2025). Suspicious emails also show a red warning banner. Gmail has a privacy tool called confidential mode, which sets an expiry date on an email and stops people from forwarding it. However, while emails are protected in transit, they are not end-to-end encrypted, unlike WhatsApp chats.

Figure 4

Google Account → Security → How you sign in to Google



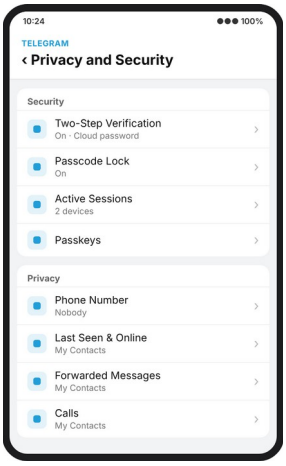
Note. Illustrative recreation of the settings screen, based on the app as of October 2026.

2.5 Telegram

For login protection, Telegram has two-step verification through a cloud password, a passcode lock and an “Active sessions” list that shows every logged-in device. Its privacy settings let users hide their phone numbers completely and connect by username. Its strongest feature is Secret Chats, which are end-to-end encrypted and can self-destruct after a set time. Unlike WhatsApp chats, ordinary Telegram chats and group chats are not end-to-end encrypted, which is a weakness. In September 2024, Telegram announced that it would give the IP addresses and phone numbers of rule-breakers to authorities if it received a valid legal request (Engadget, 2024).

Figure 5

Telegram → Settings → Privacy and Security



Note. Illustrative recreation of the settings screen, based on the app as of October 2026.

3. Additional Features Each Platform Should Add

Every platform has good security features. However, the strongest ones are mostly optional, hidden or designed for a global audience rather than for Bangladeshi users. Table 1 lists the features I propose.

Table 1

Proposed Additional Security Features

Platform	Proposed feature	Why it is needed
Facebook	A 48-hour delay before a new email, phone number or 2FA change takes effect, with alerts sent to the old contacts	Hackers and scammers change the email within minutes, so changes need to be delayed
Instagram	End-to-end encrypted DMs by default; blurring of explicit images	Sextortion often starts in DMs
WhatsApp	A Bangla warning whenever a code is requested; a mandatory two-step PIN for new accounts	Code-sharing scams are the main cause of account hijacking
Gmail	Optional end-to-end encryption; warnings for fake bKash, Nagad and bank emails	Email resets every other account, and phishing copies local brands
Telegram	End-to-end encryption for ordinary chats; a two-step verification prompt at sign-up	Most users never switch protection on

Note. Proposals are the author's own, based on the gaps identified in Section 2.

Strong security should be switched on automatically rather than left as an option for the user to activate. Most Bangladeshi users are unaware of these features, and a protection that is inactive by default fails to protect anyone because users rarely activate it.

4. Which Platform Is Most Secure?

Table 2

Security Scorecard of the Five Platforms (1 = weak, 5 = strong)

Criterion	Facebook	Instagram	WhatsApp	Gmail	Telegram
Login protection	4	4	4	5	3
Encryption by default	3	1	5	3	2
Threat detection & alerts	4	4	3	5	3
Recovery safety	2	3	3	4	2
Privacy controls	3	4	5	3	4
Total (out of 25)	16	16	20	20	14

Note. Scores are the author's assessment of the default and available settings as of October 2026.

WhatsApp and Gmail tie at 20 out of 25, but for an ordinary individual user I judge WhatsApp to be the most secure. Its strongest protection, end-to-end encryption, works from the moment the app is installed, with no action needed from the user. Gmail has the best account protection and phishing filters, and it deserves special care because it is the “master key”: whoever controls a person’s email can reset the passwords of almost everything else. Telegram scores lowest because its reputation for privacy is stronger than its defaults, while Instagram’s retreat from encrypted messaging pulls it down.

Still, no platform can protect a user who hands over a one-time code or reuses a weak password. The CCAF data shows that the typical victim is young and does not report the crime (Dhaka Tribune, 2024), which suggests that awareness matters as much as technology. In my view, the most secure account is simply the one whose owner has switched on 2FA or a passkey, checked their recovery details and learned to recognize a scam.

5. Conclusion

All five platforms offer serious security tools, but they differ in how much of that protection is on by default. WhatsApp leads for private communication and Gmail for account protection, while Telegram and Instagram leave the most content exposed. Platforms should make strong settings the default and speak to users in their own language; users, meanwhile, should spend five minutes turning on the protections that already exist.

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University of Dhaka

Department of Criminology

Professional Masters in Criminology and Criminal Justice

Course: MCCJ 527 – Organized Crime and Corruption

Group Assignment, Part I: Movie Review

Selling Bullets, Bending Rules:

Organized Crime, Corruption and the Grey Arms Market in War Dogs (2016)

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Selling Bullets, Bending Rules: Organized Crime, Corruption and the Grey Arms Market in War Dogs (2016)

1. Introduction

War Dogs (2016) is a crime comedy-drama directed by Todd Phillips (Phillips, 2016). The main characters are played by Jonah Hill and Miles Teller, who portray two young men from Miami who become arms dealers, with Ana de Armas and Bradley Cooper in supporting roles. The film was released by Warner Bros. in August 2016 and is based on a true story reported by the journalist Guy Lawson (Lawson, 2011, 2015). The two men's company, AEY, eventually wins a US Army contract worth about $300 million. The main theme revolves around how business, government and crime can become entangled with one another. The plot makes the film ideal for studying organized crime, as the criminals are disguised as ordinary businessmen.

The film raises a few critical questions about who is responsible when a crime occurs inside a legal system. It shows that a criminal enterprise does not necessarily need guns, gangs or secret meetings to cause serious harm. For criminology students, this makes *War Dogs* a useful case study to assess and test different definitions and theories. In this paper, we summarize the plot and analyze it using core course concepts.

2. Summary of the Plot

David, a struggling massage therapist in Miami, needs money because his girlfriend is pregnant. He joins his old friend Efraim, whose company bids for small military contracts online. When a weapons shipment gets stuck in Jordan, they use local fixers to deliver it to Iraq. They then meet Henry, an arms dealer on a terrorist watch list, who helps them find cheap ammunition in Albania. This deal helps AEY win a contract to supply the Afghan army.

Later on, David discovers that the ammunition was made in China, which the contract does not allow. They hide this by repackaging the bullets and supplying them anyway. After this, Efraim tries to cut Henry out of his share, which creates tension and danger for both of them. Eventually, their partnership falls apart, and the government starts an investigation. Efraim is sentenced to four years in prison, and David is sentenced to seven months of house arrest (Lawson, 2015). Throughout the film, David narrates the story, showing how an ordinary person slowly turns into a criminal. Each step seems small, but together they lead to serious fraud. The concluding scenes also show how mistrust and greed can destroy a criminal partnership.

3. Conceptual Analysis

3.1 Key Definitions

The United Nations Convention against Transnational Organized Crime (UNTOC) defines an organized criminal group as a structured group of three or more persons that exists for a period of time and acts together to commit serious crimes for financial or other material benefit (UNODC, 2004, Art. 2). A "serious crime" means any offense punishable by at least four years in prison, and no formal hierarchy is needed. Abadinsky (2013) describes organized crime through attributes such as being non-ideological and hierarchical, having limited membership, being self-perpetuating, using violence or bribery, having specialized roles, seeking monopoly and following rules.

According to Sutherland (1940), white-collar crime is a crime committed by a respectable person in the course of their occupation or by using their position. State-corporate crime refers to harm caused by government and business working together (Kramer et al., 2002). Corruption, on the other hand, is defined as the abuse of entrusted power for private gain (Transparency International, n.d.). These definitions show how organized crime can be understood through both a broad and a narrow lens. The UN definition focuses on what a group does, while Abadinsky focuses on how a group is structured. These definitions are important because the film shows a group that is clearly criminal but does not look like a traditional gang. Using both lenses helps us judge AEY more fairly in the next sections.

3.2 Classification: Is AEY "Organized Crime"?

Under the UNTOC framework, this is a case of organized crime, as Efraim, David, the financier Ralph, Henry and the Albanian suppliers formed a group of more than three people who worked together over several years and engaged in fraud and conspiracy with the objective of earning profit. It is also transnational, involving the United States, Jordan, Iraq, Albania and Afghanistan. However, the case does not match Abadinsky's (2013) attributes, as there were no rituals or exclusive membership, no systematic use of violence, and the organization collapsed once its leaders were arrested.

Therefore, AEY is best classified as organized white-collar crime, where a legal company uses legal channels to commit organized fraud. From the outside, AEY did not look criminal, so nobody checked it. This classification is important because it challenges the conventional idea that organized crime is always violent. Many organized crime groups function through companies, contracts and bank accounts rather than weapons and threats. AEY is a classic example of how crime can hide inside ordinary business activity.

3.3 Models of Organized Crime

The course introduced four main models for explaining how organized crime is structured. Table 1 summarizes how well each one explains the events of the film.

Table 1

Fit Between Organized Crime Models and the Events of War Dogs

Model	Core idea	Fit with the film
Hierarchical/bureaucratic (Cressey, 1969)	Mafia-style bureaucracy with ranks	Weak fit: AEY had no ranks or code
Patron–client (Albini, 1971)	Patrons give access in return for loyalty and money	Partial fit: Henry controlled access to the supplies
Enterprise (Smith, 1980)	Crime as a business, ranging from legal to illegal	Strong fit: AEY moved from legal bidding to fraud
Network (Morselli, 2009)	Brokers linking separate groups hold power	Strong fit: Efraim and Henry connected the Pentagon to the Balkan supply

Note. Fit ratings are the group's assessment.

The enterprise model suits this case best, as the US government created the demand and outsourced it online (Smith, 1980). AEY drifted into illegality when the profits were high and the risk of getting caught was low. The case also fits the network model, as Efraim and Henry connected the whole business, from the Pentagon down to the Balkan supply. In this model, criminals act as nodes in a network, and those who connect different parts of it hold the most power (Morselli, 2009). The patron–client model fits only partly, as Henry acted like a patron who controlled access to the supplies in return for a share of the profit (Albini, 1971). On the other hand, the hierarchical model fits poorly, because AEY had no ranks, rules or code, and everything depended on Efraim (Cressey, 1969).

Taken together, the models show that AEY was not a fixed organization but a flexible business network. Its members joined and left depending on the deal, which is typical of modern criminal enterprises. This flexibility made AEY harder to detect than a traditional criminal group.

3.4 Theories of Offending and Corruption

The whole scenario can be explained through the lens of different criminological theories. The first is rational choice theory, which explains that an offender commits a crime after weighing the profit against the risk of punishment or of getting caught (Cornish & Clarke, 1986). Here, the group hid the Chinese origin of the ammunition because the profit was huge and the risk seemed small, a situation that situational crime prevention could have addressed. Another relevant theory is routine activity theory, which explains that a crime occurs when a motivated

offender, a suitable target and the absence of a capable guardian converge at the same time and place (Cohen & Felson, 1979). The plot shows how a poorly monitored contract system gave the offenders a suitable target to exploit and earn huge profits from.

Strain theory explains crime as a result of economic strain: people in society share the same goals, but the opportunities to reach those goals are unequal, which creates a strain that can lead to crime (Merton, 1938). In this case, it explains David's situation, as he needed money but did not have a legitimate way to earn it, so after finding no other way, he joined the group. It also reflects institutional anomie, where the desire to earn money overrides other institutional values in society (Messner & Rosenfeld, 1994). In the film, this is seen through David, who continuously lies to his partner, Iz, about his illegally earned money. His situation shows how financial pressure can push an ordinary person towards crime.

Sutherland's (1947) differential association theory is also very relevant here. It explains that criminal behavior is learned through intimate relationships, and that this learning varies in frequency, duration, priority and intensity. Here, David learned criminal methods and excuses from Efraim. The characters also use techniques of neutralization to justify their actions; for example, they suggest that no one is harmed because the soldiers still receive working bullets, which is a form of denial of injury (Sykes & Matza, 1957).

Klitgaard's (1988) formula for corruption, Corruption = Monopoly + Discretion − Accountability, also describes the scenario. Here, Henry held a monopoly over the supplies, officials and fixers had the discretion to decide freely, and accountability was low during wartime.

No single theory fully explains the characters' behavior. Instead, the theories work together: strain creates the motive, differential association teaches the methods and neutralization removes the guilt. Weak oversight and corruption additionally provide the opportunity. This combination shows why both individual and structural explanations are needed.

3.5 The Transnational Dimension and the Licit–Illicit Nexus

The case shows how legal and illegal markets connect across national borders. The ammunition starts as legal surplus in Albania, passes through brokers and companies, and arrives in Afghanistan as a legal US government purchase. The only hidden and illegal element is its origin, which makes the process similar to money laundering. Henry was on a terrorist watch list, but he used AEY's clean image to stay in the market, which illustrates the crime–terror nexus, the link between organized crime and terrorism. Post-communist states with old stockpiles and weak institutions, such as Albania, can easily become supply hubs.

The case also shows the limits of national law enforcement. Each nation can monitor only one part of the network, so no one can see the whole picture. International cooperation and information sharing are therefore essential parts of prevention.

4. Strengths and Weaknesses of the Film

The film breaks the mafia stereotype by displaying crime as a pure business with websites and invoices. It shows how a small business can exploit a government system to earn millions. It fits the enterprise and network models more closely than films such as *The Godfather*. The characters never view themselves as criminals, which is psychologically realistic. The core facts also remain true, including the contract, the repackaged Chinese ammunition and the sentences (Lawson, 2015).

However, the film's comedic elements glamorize the money and make fraud look harmless. Henry is a composite character, and the kidnapping-style threats in Albania did not happen as shown. The film focuses too much on two individuals rather than on the system. It also leaves out the Gërdec depot explosion in Albania, which killed 26 people in 2008 (Al Jazeera, 2008). Nor does it explain how a company run by a man in his early twenties passed the Pentagon's vetting. Overall, the film is a useful starting point for discussion, but it should not be treated as a documentary. Viewers need to compare it with real reports to understand the full story. Its entertainment value is high, but its educational value depends on how critically it is watched. In short, the portrayal is conceptually accurate but exaggerated in style.

5. Conclusion

AEY meets the UNTOC definition of an organized criminal group but does not fit the traditional mafia model. It is best explained by the enterprise and network models. The behavior of its members is also clearly explained by rational choice theory, strain theory and differential association theory. The film is accurate in substance, but its style exaggerates the story, so viewers need a critical mindset.

In the Bangladeshi context, public procurement also relies on middlemen. Authorities should therefore vet contractors, disclose who really owns companies and check the origin of goods. In future, criminologists should pay more attention to crimes hidden inside legal businesses and government contracts.

As a film, *War Dogs* is entertaining as well as concerning. It shows that organized crime can look completely normal from the outside. Overall, our group found that the film is a strong example of how organized crime is changing. Modern organized crime does not always look like the mafia; it can look like a small company with a laptop and a government contract. Studying this film helped us connect the theories from our course to a real case.

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